Physics-Based Core Compact Model of 2D MoS_2 FET Considering Fermi-Dirac Statistics

Swapna Sarker, Ankur Garg, Prabhat Ranjan, Abhishek Kumar, and Avirup Dasgupta

IIT Roorkee, Uttarakhand, India (Email:swapna_s@ece.iitr.ac.in, avirup@ece.iitr.ac.in, 7386842881)

2024 8th IEEE Electron Devices Technology & Manufacturing Conference (EDTM) | 979-8-3503-7152-9/24/$31.00 ©2024 IEEE | DOI: 10.1109/EDTM58488.2024.10511512

Abstract

In this paper, we present a physics-based core compact model for two-dimensional (2D) molybdenum disulfide (MoS_2) channel material-based FET. The FET characteristics have been studies using ab-initio simulations as well as experimental data. The model is derived from closed-form solutions of Poisson's equation considering 2D density of states and Fermi-Dirac statistics along with drift-diffusion transport model. The proposed model has been implemented in Verilog-A and validated with ab-initio simulations and experimental data.

Keywords: 2D, TMD, Fermi-Dirac Statistics, ATK, DFT, Compact Model

Introduction

In the post-silicon fusion era 2D transition metal dichalcogenides (TMDs) have recently emerged as an appealing alternative channel material for atomically thin field-effect transistors (FETs) [1]–[5]. TMDs, such as layered Molybdenum disulfide (MoS_2) [Fig. 1(a)], are a desirable choice for channel material in scaled devices for low-power electronics and other applications because of impressive electrostatic control and favorable material properties [Fig. 1(b)] [1].

A precise physics-based core compact model for 2D FETs that (a) takes into account 2D density of states, (b) captures appropriate Fermi-Dirac statistics, (c) addresses MoS_2 properties from the standpoint of the material, (d) captures length dependent quasi-ballistic transport and (e) is compatible with industry standard SPICE simulators is of utmost importance and immediate requirement to explore optimal circuit simulations. However, a single model ensuring all of the requirements mentioned above has not been reported yet [1]–[5].

In this paper we develop a physics-based core compact model for layered MoS_2 channel material-based FET. In the model development, 2D density of states, Fermi-Dirac statistics, and unique properties of MoS_2 have been taken into account to calculate core surface potential. This is then used to compute the current's drift-diffusive components while accounting for various second-order effects including the quasi-ballistic regime.

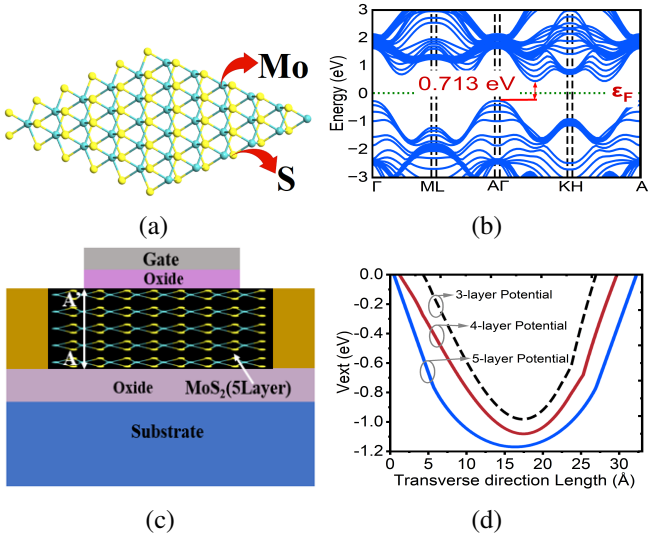

Fig. 1: ((a) Atomistic-configuration of monolayer MoS_2 material, (b) Electronic band-structure of 5-layer MoS_2 calculated using DFT, (c) Schematic of the 5-layered MoS_2 channel material-based DG FET, and (d) Potential for distinct MoS_2 layers along the cut line AA' of (c), where all corresponding layers (3 layer, 4-layer, and 5 layer) exhibit nearly smooth potentials.

Model Description

In this section, first, we proceed by studying the 2D material property through first-principle Density Functional Theory (DFT) simulations (ATK) [6]. Next, a core surface potential based model is derived for thin body MoS_2-based symmetric double-gate (DG) FET.

A. DFT

We have employed DFT to simulate layered MoS_2 with the objective to study the crititcal material properties that would influence FET behavior and also look at the potential distribution under field [6]. A schematic representation of the atomistic configuration is shown in Fig. 1(a), where the unit cell comprises three atoms, forming a hexagonal lattice structure in accordance with the P63/mmc (194) space group. A thin body in 3, 4 and 5-layer MOS_2 channel material-based FET leads to almost smooth potential along the transverse direction [Fig. 1(c) (A-A' cutline)] of the device as shown in Fig. 1(d). This allows us to solve the Poisson equation assuming a continuous potential profile as described in the next section. Other results from DFT simulations have also been used in the model development and validation process.

Authorized licensed use limited to: FUDAN UNIVERSITY. Downloaded on February 16,2026 at 06:47:06 UTC from IEEE Xplore. Restrictions apply.

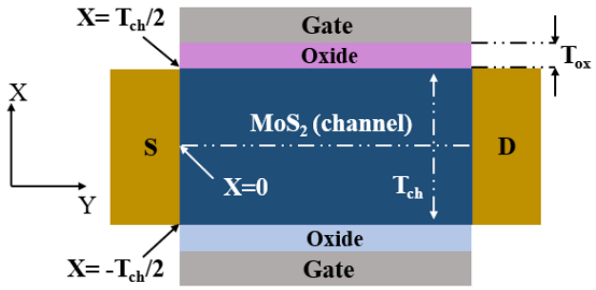

Fig. 2: Cross-section view of MoS_2 channel symmetric Double Gate FET. In the schematic, T_{ch} and t_{ox} represent the channel thickness and oxide thickness respectively.

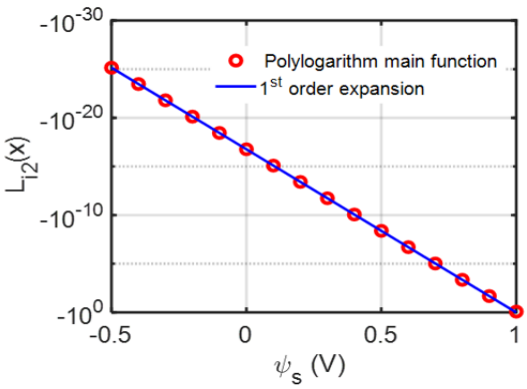

Fig. 3: Plot of Li_2 and the 1st order approximation with respect to potential ψ_s. The 1st order approximation works well for the entire bias range of interest.

B. Core Model: Surface Potential Formulation

Using the coordinate systems and potential as shown in Fig. 2, Poisson's equation for the MoS_2-based channel region with only mobile carriers (electrons) of a symmetric double-gate MOSFET can be written as [7]

$$\frac{d^2\psi}{dx^2} = \frac{qD\phi_t}{\epsilon_{ch}T_{ch}} \ln(1 + e^{\frac{\psi_s}{\phi_t}}) \tag{1}$$

where ψ denotes the electrical potential along the x-direction at any given position in the y-direction, q is the electronic charge, D is the 2D density of states [5], ϵ_{ch} is the permittivity, T_{ch} is the thickness of the channel and ϕ_t is the thermal voltage. A key novelty of this model is that we have employed Fermi-Dirac statistics instead of Boltzmann approximation to model the carrier density. Integrating (1) gives

$$\left(\frac{d\psi}{dx}\right)^2 = \frac{2\phi_t D}{\epsilon_{ch}T_{ch}}\left[-\phi_t Li_2\left(-e^{\frac{\psi_s}{\phi_t}}\right)\right]\Big|_{\psi_0}^{\psi} \tag{2}$$

where, Li_2 is the 2nd order polylogarithm function. For the range of biases for MoS_2 FETs, we can approximate Li_2 with the first order term, as shown in Fig. 3. Using this approximation allows us to integrate (2)

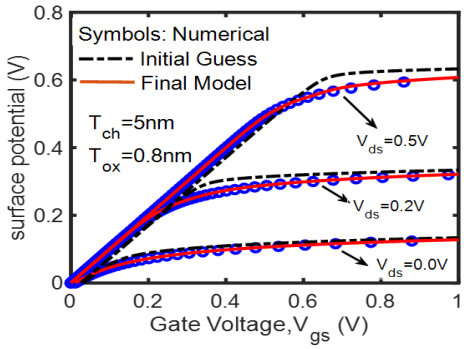

Fig. 4: Variation of surface potential ψ_s with respect to gate voltage V_{gs} for different V_{ds} values. The initial guess for surface potential is illustrated by the black dashed lines. The red solid lines indicate the surface potential's final value after being corrected in one iteration using Halley's method from the initial estimate.

using the boundary condition at the interface $(\frac{d\psi}{dx})_{x=0}$ =0 to get

$$\frac{d\psi}{dx} = \sqrt{\frac{2\phi_t^2 D}{\epsilon_{ch}T_{ch}}\left[e^{\frac{\psi_s}{\Phi_t}} - e^{\frac{\psi_0}{\phi_t}}\right]} \tag{3}$$

where, ψ_0= $\psi(x = 0)$ is the potential at the center of the 5-layer MoS_2 film. Integrating once again, and using boundary condition $x = T_{ch}/2$ at the interface [7], the surface potentials ψ_s can be described as

$$\psi_s = \psi_0 + \phi_t ln\left[1 + \frac{DT_{ch}}{8\epsilon_{ch}}e^{\frac{\psi_0}{\phi_t}}\right] \tag{4}$$

Employing the boundary condition that the electric displacement will be continuous at the oxide-2D material interface, we can further write

$$C_{ox}(V_{gs} - V_{fb} - \psi_s) = \epsilon_{ch}\left(\frac{d\psi}{dx}\right)_{x=T_{ch}/2} \tag{5}$$

where V_{fb} is the flat band voltage and $C_{ox} = \epsilon_{ox}/t_{ox}$. Substituting the value of $\frac{d\psi}{dx}$ and ψ_s from (3) and (4) respectively in (5) we can write

$$V_{gs} - V_{fb} = \psi_0 + \phi_t ln\left[1 + \frac{DT_{ch}}{8\epsilon_{ch}}e^{\frac{\psi_0}{\phi_t}}\right] + \frac{\phi_t D}{2C_{ox}}e^{\frac{\psi_0}{\phi_t}} \tag{6}$$

In (6), the first and second terms dominate in weak inversion and the third term is for the strong inversion. Eq. (6) is solved using a robust initial guess for ψ_0 followed by a single iteration using Halley's method. The initial guess is given as

$$\psi_0 = min\{\psi_{0WI}, \psi_{0SI}, C\} \tag{7}$$

where C is a smoothing parameter while ψ_{0WI} and ψ_{0SI} are the approximate surface potentials in weak and strong inversion, respectively, given as

$$\psi_{0WI} = \frac{Vgs - Vfb}{1 + \frac{DT_{ch}}{8\epsilon_{ch}}} - \phi_t \tag{8}$$

Authorized licensed use limited to: FUDAN UNIVERSITY. Downloaded on February 16,2026 at 06:47:06 UTC from IEEE Xplore. Restrictions apply.

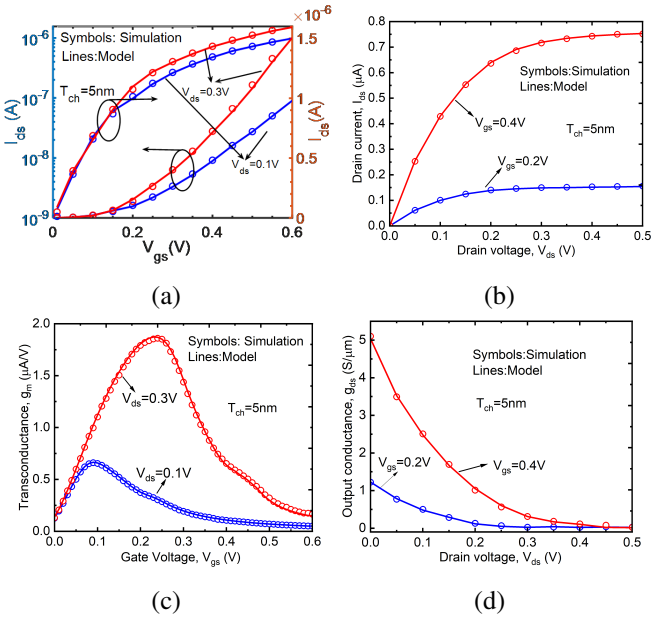

Fig. 5: Validation of proposed model's (a) transfer characteristics (I_{ds}-V_{gs}) for different V_{ds}, (b) output characteristics (I_{ds}-V_{ds}) for different V_{gs} (c) Transconductance, and (d) output conductance against atomistic simulation data. The proposed MoS_2 channel-based SDG FET model exhibits great agreement with the simulation data.

$$\psi_{0SI} = \phi_t \ln\left(\frac{2C_{ox}V_{gsfb}}{D\phi_t}\right) \tag{9}$$

As illustrated in Fig. 4, the surface potential model shows excellent accuracy throughout the bias range.

C. Core Model: Drain Current

The core drain current (I_{ds}) considering 2D density of states (DOS), quasi-ballistic framework (covering drift-diffusive to full ballistic) and current continuity as well as various second order effects can be expressed as [8]

$$I_{ds} = \frac{\mu_{qbeff}C_{gg}}{\sqrt{1+\theta_{sat}^2\psi_{ds}^2}}\frac{W}{L}(V_{go}-\psi_m-V_{th})\psi_{ds}(1+\lambda V_{ds}) \tag{10}$$

where, W and L are the width and length respectively, C_{gg} is the gate capacitance per unit area. θ_{sat}, and λ are the coefficients for velocity saturation, and channel length modulation respectively [8]. $V_{go} = V_{gs} - V_{OFF}$ with V_{OFF} is cut-off voltage. Also, ψ_m=$(\psi_{ss} + \psi_{sd})/2$ and $\psi_{ds} = \psi_{sd} - \psi_{ss}$ with ψ_{ss} and ψ_{sd} denoting the source and drain side surface potential, respectively. μ_{qbeff} is the effective carrier mobility considering both drift diffusive mobility as well as length-dependent ballistic mobility [9].

Validation

Fig. 5 shows the validation of the proposed model with the atomistic simulation for different biases. Not only the predicted current (Fig. 5 (a) and (b))but the derivatives in the form of transconductance and output conductance (Fig. 5(c) and (d)) also show good agreement.

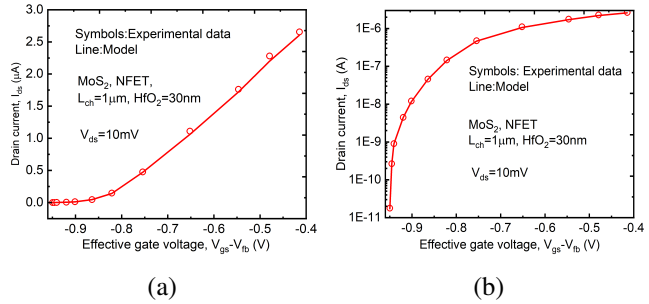

Fig. 6: Validation of drain current I_{ds} with respect to effective gate voltage (V_{gs}-V_{fb}) of proposed MoS_2 channel DG FET in (a) linear, and (b) logarithmic scale against the experimental data reported in [10]. The model shows excellent agreement with the experimental data.

The model has also been validated against the experimentally reported MoS_2-based DG-FET [10], as illustrated in Fig. 6, showing excellent accuracy.

Conclusion

A surface potential and physics-based core compact model for MoS_2 FET has been successfully developed and thoroughly validated by detailed atomistic simulations as well as experimental data. By incorporating 2D density of states, Fermi-Dirac statistics, quasi-ballistic transport as well as various second order effects such as mobility degradation, velocity saturation, and channel length modulation, the proposed SPICE compatible model serves as a reliable tool to evaluate 2D FET in circuit simulations.

Acknowledgment

We acknowledge the support from IIT Roorkee; SERB, Govt. of India; and UC Berkeley.

References

[1] Cao et al. *IEEE TED*, vol 61, no. 12, 2014.
[2] Cao et al. *IEEE TED*, vol 61, no. 10, 2014.
[3] S. A. Ahsan et al. *IEEE TED*, vol. 68, no. 6, June 2021.
[4] L. Wang et.al., *IEDM*, 2018, DOI: 10.1109/IEDM.2018.8614655.
[5] C. Yadav et. al. *IEEE TED*, vol. 64, no. 3, March 2017.
[6] D. Marian et.al. *IEDM*, 2016, doi: 10.1109/IEDM.2016.7838413.
[7] Y. Taur et. al. *IEEE EDL*, vol. 21, no. 5, May 2000.
[8] S. Khandelwal et al. *IEEE TED*, vol. 60, no. 10, Oct. 2013.
[9] J. Wang et. al. *IEEE TED*, vol. 50, no. 7, July 2003.
[10] S. Bhattacharjee et. al. *IEEE Transactions on Nanotechnology*, vol. 18, 2019.
[11] Nan Ma and Debdeep Jena 2015 *2D Mater*. 2 015003.

Authorized licensed use limited to: FUDAN UNIVERSITY. Downloaded on February 16,2026 at 06:47:06 UTC from IEEE Xplore. Restrictions apply.